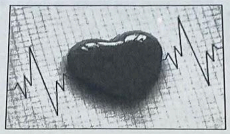
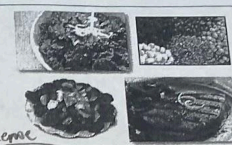
Salt, Seafood, Milk, Some Vegetables

iodine



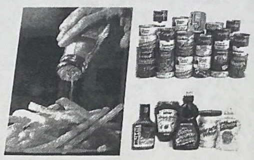
Heartbeat Regulation, Anti-Hypertensive

potassium
calcium



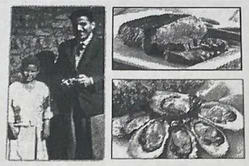
heme
Meat, Fish, Poultry, Beans, Breakfast Cereal

iron



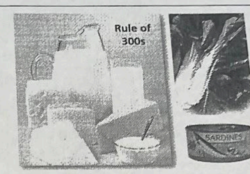
Electrolyte, Canned Goods, Condiments

sodium



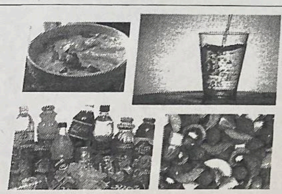
Hypogonadism (Def), Oysters, Meat

zinc



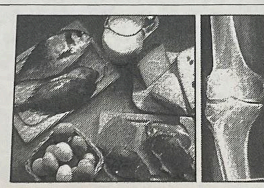
1 Serving Milk, Yogurt, Cheese ~ 300 mg

calcium



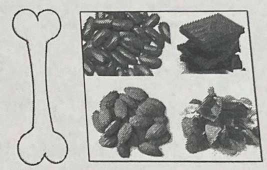
Nutrient Transport, Hydration

water



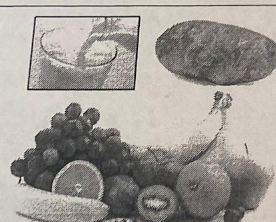
Bone Health, Needed to Make ATP (Energy)

phosphorus



Bone Health, Anti-Hypertensive, Nuts, Beans

magnesium

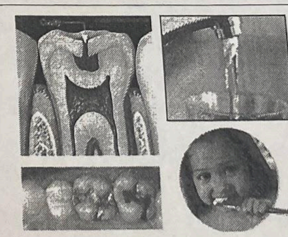


Electrolyte, 4,700 mg/day, Fruit/Veg/Dairy

potassium

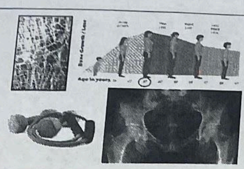
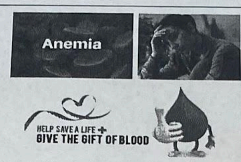
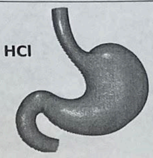
Soy Sauce, Salt (NaCl), Processed Foods

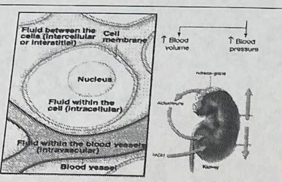
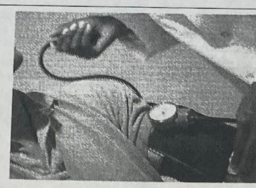
chloride



Bones & Teeth, Toothpaste, Drinking Water

fluoride

		
Anti-Hypertensive, Osteoporosis (Deficiency)	Hemoglobin, Anemia (Def), 8 to 18 mg/day	Electrolyte, Hydrochloric Acid (for Digestion)
Calcium	iron	chloride

		
Blood Pressure & Body Temp Regulation	Hypertension, Hyponatremia (Deficiency)	Goiter (Def + Tox), Cretinism (Deficiency)
water	sodium	iodine

Increase (I) or Decrease (D)?

1. Drinking a gallon of water at one time will __↓__ the concentration of sodium in the blood, resulting in hyponatremia.
2. The DASH (Dietary Approaches to Stop Hypertension) will likely __↓__ blood pressure.
3. Calcitonin __↑__ calcium deposition on the bone to __↓__ blood calcium after a calcium-rich meal.
4. PTH and Vitamin D __↑__ calcium absorption in the intestines to __↑__ blood calcium levels.
5. Vasoconstriction __↓__ the diameter of a blood vessel to __↑__ blood pressure.
6. Heavy alcohol consumption will likely __↓__ bone mineral density.

High or Low?

1. A blood pressure of 140/90 H
2. The DASH diet is __L__ in sodium.
3. Joey regularly consumes 3200 mg of potassium. His intake is __L__ (rec = 4700)
4. A pre-menopausal woman consumes 10 mg of iron daily. Her intake is __L__ (rec = 18)

What nutrients contribute to bone health?

Vitamins

- vit. D
- vit. A
- vit. K

Minerals

- Calcium
- magnesium
- phosphorus

What minerals serve as electrolytes?

- sodium
- potassium
- chloride

Which one is the main intracellular cation?

potassium

hemolytic anemia = vitamin E deficiency

macrolytic = B12 or folate

microlytic = iron

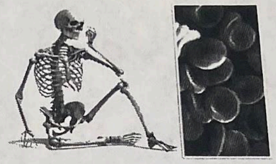
sickle cell =

Rickets, Osteomalacia, Osteoporosis

Vitamin D

Fruit (Melons, Avocados), Vegetables, Meat

vitamin B6

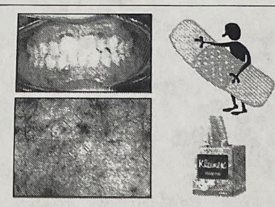


Bone Health, Blood Clotting

Vitamin K

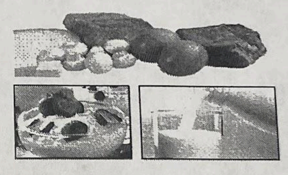
Whole Grains, Mushrooms, Yogurt, Seeds

pandothenic acid



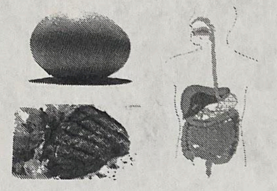
Scurvy, Wound Healing, 90 mg/d or 75 mg/d

Vitamin C



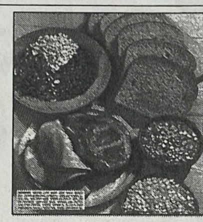
Milk, Yogurt, Fish, Eggs

vitamin D



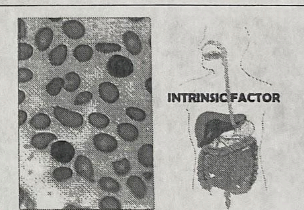
Eggs (Yolks), Meat, GI Tract, Mushrooms

biotin



Pork (Ham), Whole Grains, Enriched Foods

thiamin

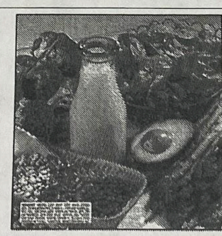


Myelin Sheath, Anemia/Nerve Damage (Def)

vitamin B12

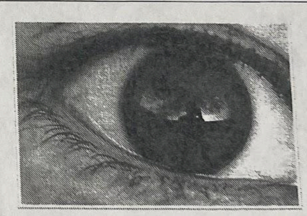
Green Leafy Vegetables, GI Tract

vitamin K



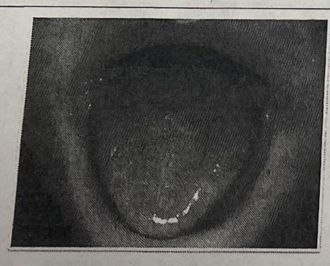
400 mcg/day, Green Veggies, Prenatal Vits

folate



Vision, Xerophthalmia (Blindness)

Vitamin A



Ariboflavinosis (Deficiency)

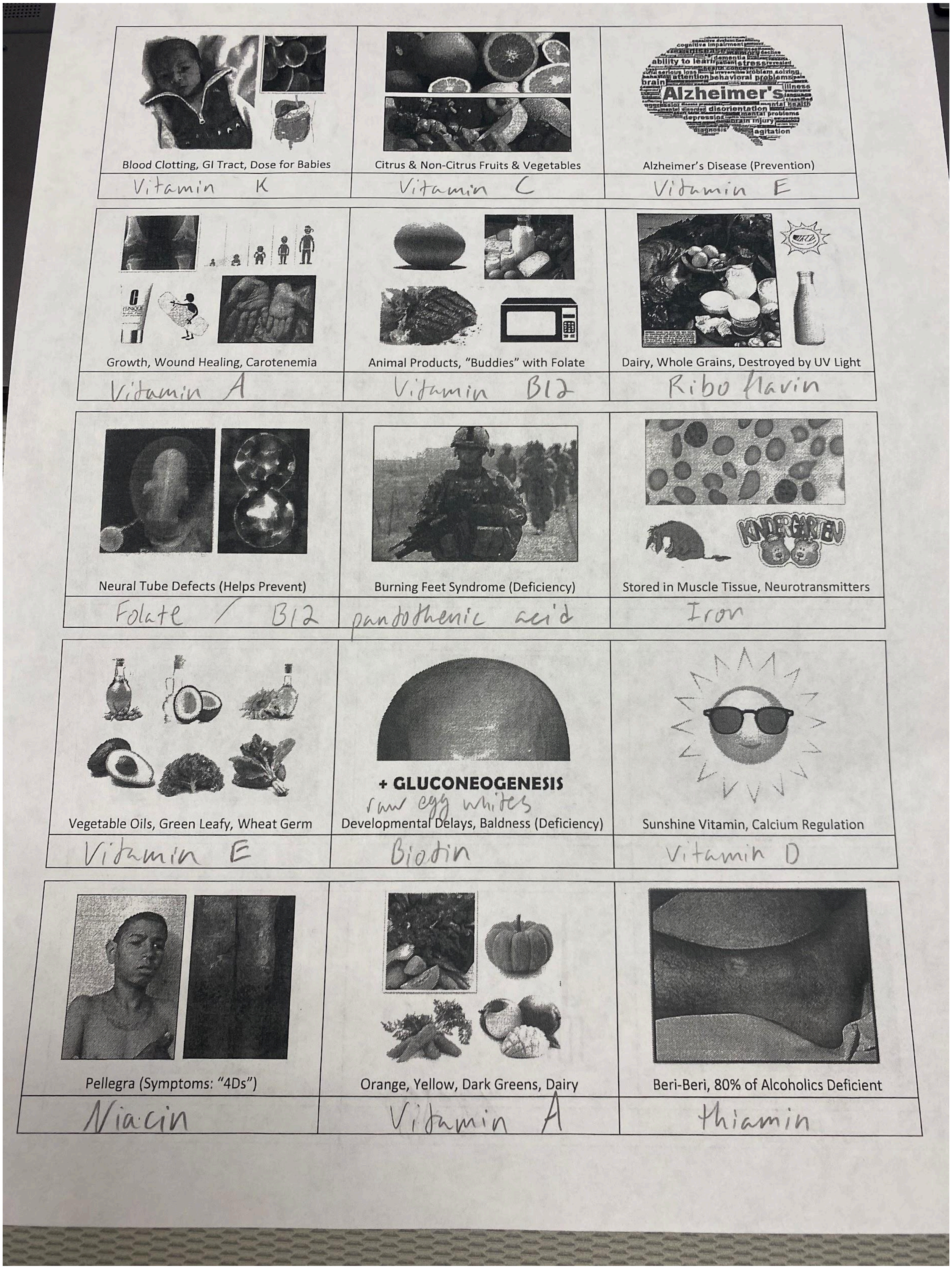
Riboflavin

Wound Healing, Antioxidant, Hemolysis

vitamin E

Meat, Whole Grains, Enriched Products

Niacin



Blood Clotting, GI Tract, Dose for Babies	Citrus & Non-Citrus Fruits & Vegetables	Alzheimer's Disease (Prevention)
Vitamin K	Vitamin C	Vitamin E
Growth, Wound Healing, Carotenemia	Animal Products, "Buddies" with Folate	Dairy, Whole Grains, Destroyed by UV Light
Vitamin A	Vitamin B12	Ribo flavin
Neural Tube Defects (Helps Prevent)	Burning Feet Syndrome (Deficiency)	Stored in Muscle Tissue, Neurotransmitters
Folate / B12	pantothenic acid	Iron
Vegetable Oils, Green Leafy, Wheat Germ	+ GLUCONEOGENESIS raw egg whites Developmental Delays, Baldness (Deficiency)	Sunshine Vitamin, Calcium Regulation
Vitamin E	Biotin	Vitamin D
Pellegra (Symptoms: "4Ds")	Orange, Yellow, Dark Greens, Dairy	Beri-Beri, 80% of Alcoholics Deficient
Niacin	Vitamin A	thiamin

Electrolytes, required for fluid balance = sodium, chloride, potassium

Can be synthesized in the body = niacin, biotin, vit K, vit D
Beriberi (deficiency) = thiamin/B1
Maintains myelin sheath around nerve cells = vit B12
Broccoli, potatoes, kiwi, strawberries, peppers = vit C

Hyponatremia = sodium deficiency
Standard nutrients added to enriched foods = Thiamin, Riboflavin, Niacin, Folate, Iron